Customer Churn Predictor

Methodology Note

Goal: Flag customers who may be about to leave, so the team can contact them before they do.

Data: A sample dataset of 200 customers, with each customer's last order date, orders in the last 30 days, orders in the previous 30 days, and total spend.

Churn signals used:

1. **Inactive 30+ days:** no order in the last 30 days or more.
2. **Order frequency drop:** orders in the last 30 days are 50% or less of the previous 30 days.
3. **High-value customer gone quiet:** top 25% by total spend and no order in 20+ days. This is a priority signal because losing these customers costs the most.

Scoring: Each customer gets 1 point per signal (0 to 3). Anyone scoring 1 or more is flagged. The list is ranked by score, with ties broken by total spend, so the most valuable at-risk customers come first.

Result: 123 of 200 customers (61.5%) were flagged. 9 customers scored 3 and should be contacted first, led by C1102, C1068 and C1097, all with spend above $4,800.

Limitations and next steps: The thresholds (30 days, 50%, 20 days, top 25%) are rules of thumb, not learned from real churn history, and the data is simulated. Flagging 61.5% of customers is too broad to act on in practice, so a real team would contact only the score-2 and score-3 groups first, or raise the thresholds. With real data, I would validate the thresholds against customers who actually left, or train a logistic regression to weight the signals.